

SOFTWARE REQUIREMENTS SPECIFICATION

STM32F103 Bare-Metal Environmental Temperature Data Logger

Document	STM32F103 Bare-Metal Environmental Temperature Data Logger SRS	Status	Project Technical Specification
Implementation	Bare-metal C / register-level firmware	Primary MCU	STM32F103 / Cortex-M3
Storage	W25Q128 128-Mbit SPI Flash	UI	SSD1306 128x32 + 4x4 keypad

1. System Purpose and Scope

The system acquires an NTC thermistor signal through ADC1, filters the ADC result, calculates temperature using the NTC Beta model, associates samples with a TIM3-based system-time counter, buffers records in RAM, and stores completed records in a W25Q128 SPI Flash. A 128x32 SSD1306 OLED provides live and historical information, while a 4x4 keypad provides user input. A three-state LED alert interface indicates normal, high, and low temperature conditions.

2. System Architecture and Data Flow

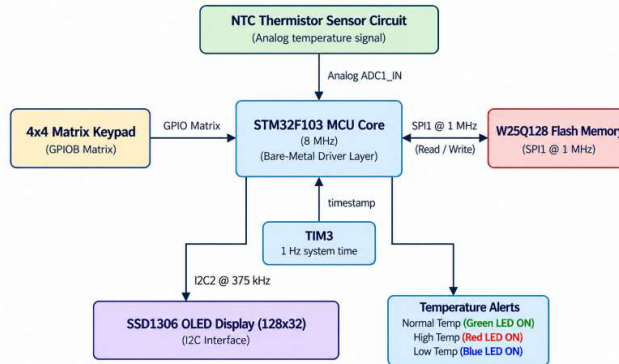


Figure 1. Functional architecture: sensor acquisition, processing, timestamping, buffering, non-volatile storage, user interface, and temperature alerts.

2.1 Operating Boundaries

Item	Requirement / Implementation Boundary
ADC acquisition	ADC1 EOC interrupt is implemented.
System time	TIM3 uses PSC=799 and ARR=9999; under the 8 MHz timer-clock assumption used by the code, the interrupt rate is 1 Hz and increments system seconds.
Flash interface	Standard single-bit SPI commands are used.
Execution model	Acquisition is interrupt-assisted; SPI Flash and I2C routines contain polling loops

3. Functional Requirements

3.1 Temperature Acquisition and Calculation

REQ-F-01 — ADC1 shall acquire the NTC analog input on PA0 and signal conversion completion through the ADC interrupt handler.

REQ-F-02 — The acquisition path shall apply the integer EMA filter: `adc_filtered = ((adc_filtered * 15) + adc_raw) >> 4`; the first sample initializes the filtered value.

REQ-F-03 — Temperature processing shall convert ADC counts using 3.3 V and 4095 counts, calculate NTC resistance using $R_{\text{fixed}} = 9800\ \Omega$, $R_0 = 10000\ \Omega$, $T_0 = 298.15\ \text{K}$, $B = 3950$, and evaluate the Beta equation using the implemented custom logarithm routine.

3.2 Timestamping and RAM Buffering

REQ-F-04 — TIM3 shall provide the system-time base used to generate `uint32_t` timestamps. With PSC=799 and ARR=9999 under the stated 8 MHz timer-clock assumption, the timer interrupt is 1 Hz.

REQ-F-05 — Two ping-pong RAM buffers shall be maintained. Each buffer contains 64 `uint32_t` timestamps and 64 float temperatures; timestamp and temperature storage is 512 bytes per buffer.

REQ-F-06 — When the active buffer reaches 64 samples, its write flag shall be set and logging shall switch to the alternate buffer.

REQ-F-07 — A completed buffer shall be committed to Flash using two 256-byte page-program operations: one timestamp page and one temperature page.

3.3 Flash Storage and Metadata Integrity

REQ-F-08 — The W25Q128 driver shall support JEDEC ID, Read Data, Page Program, Write Enable, Status Register, and Sector Erase operations.

REQ-F-09 — Metadata shall contain a magic value and a 32-bit XOR checksum calculated over the first seven `uint32_t` fields, excluding the checksum field.

REQ-F-10 — Metadata records shall be journaled across two 4 KB sectors at 0x030000 and 0x031000. When the active metadata sector is full, the alternate sector shall be erased and used as the next target.

REQ-F-11 — On startup, the logger shall scan both metadata sectors, select the highest valid sequence number, and restore the timestamp and Flash write pointers.

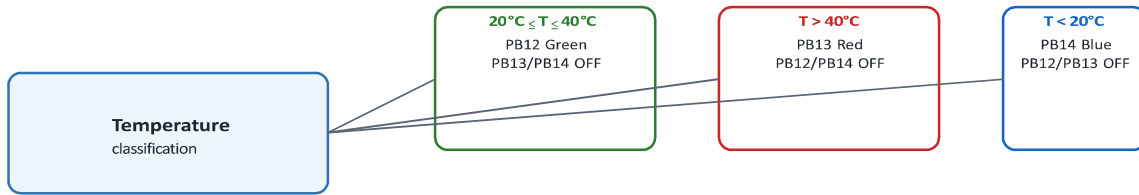
3.4 User Interface and Historical Data

REQ-F-12 — The 4x4 keypad shall be scanned as a GPIO matrix using rows PB0, PB1, PB4, PB5 and columns PB6–PB9. A key-release state shall prevent repeated registration of a held key.

REQ-F-13 — The SSD1306 shall use a 128x32-pixel framebuffer of 512 bytes with custom 5x7 font rendering; the framebuffer shall be transmitted over I2C2 to address 0x3C.

REQ-F-14 — Historical mode shall read timestamp and temperature data from Flash in 64-sample windows and provide page navigation, timestamp range, average temperature, and navigation status.

3.5 Temperature Alert System



Thresholds are mutually exclusive; the normal range includes 20°C and 40°C.

Figure 2. Alert-state logic. Exactly one LED state is active for a valid temperature classification; the other alert outputs remain OFF.

Temperature condition	GPIO	Indicator state	Other alert LEDs
$20^{\circ}\text{C} \leq T \leq 40^{\circ}\text{C}$	PB12	Green LED ON	PB13/PB14 OFF
$T > 40^{\circ}\text{C}$	PB13	Red LED ON	PB12/PB14 OFF
$T < 20^{\circ}\text{C}$	PB14	Blue LED ON	PB12/PB13 OFF

4. Hardware Interfaces and Resource Specification

Interface	Pins / Device	Configuration / Verified Value
MCU	STM32F103 / Cortex-M3	Bare-metal register-level C; code calculations use an 8 MHz clock assumption.
ADC1	PA0 / ADC channel 0	12-bit ADC, EOC interrupt. Continuous ADC mode is commented out in the supplied source.
SPI1	PA4 NSS, PA5 SCK, PA6 MISO, PA7 MOSI	Master, CPOL=0, CPHA=0, 8-bit, software NSS. Measured SCK = 1 MHz.
I2C2	PB10 SCL, PB11 SDA; SSD1306 0x3C	Master; Fast-Mode intent. Measured SCL $\approx$ 375 kHz.
TIM3	APB1 timer	PSC=799, ARR=9999; 1 Hz interrupt under the 8 MHz timer-clock assumption.
Keypad	Rows PB0, PB1, PB4, PB5; columns PB6–PB9	4x4 matrix scan; columns driven low, rows read with pull-ups; key-release state.
Alert LEDs	PB12 / PB13 / PB14	Green = normal, Red = high, Blue = low; remaining outputs OFF.

5. Software Architecture

Layer	Modules / Responsibility
Application	Temperature processing, data logger, historical-data UI, keypad handling, alert-state handling.
Device drivers	W25Q128 Flash, SSD1306 OLED, SPI1, I2C2, ADC1, TIM3, GPIO.
Register level	RCC, GPIO, SPI, I2C, ADC, TIM and NVIC registers accessed directly.
Hardware	STM32F103, NTC circuit, W25Q128, SSD1306, 4x4 keypad, three alert LEDs.

6. Flash Organization and Memory Requirement

Item	Implementation
W25Q128	128-Mbit external SPI Flash; standard single-bit SPI command interface.
Page Program	Command 0x02; maximum payload is 256 bytes per page.
Read Data	Command 0x03; used for historical data and metadata recovery.
Sector Erase	Command 0x20; used for 4 KB metadata-sector rotation.
Metadata sectors	0x030000 and 0x031000; sequence + magic + XOR checksum.
Ping-pong buffer	64 timestamps + 64 temperatures per buffer = 512 bytes; two buffers = 1024 bytes.
OLED framebuffer	$128 \times 32 / 8 = 512$ bytes.

7. Data and Control Flow

Step	Input	Processing	Output
1	NTC analog signal	ADC1 EOC interrupt	Raw ADC value
2	Raw ADC	EMA filtering	Filtered ADC
3	Filtered ADC	Voltage → resistance → custom ln → Beta equation	Temperature °C
4	Temperature + TIM3	Attach current system seconds	Timestamped sample
5	Timestamp + temperature	Append to active A/B buffer	64-sample completed buffer
6	Completed buffer	Two page-program operations	W25Q128 data pages
7	Flash state	Metadata sequence/checksum update	Recoverable logger state
8	Keypad + Flash data	UI processing	OLED live/history display + alert status

8. Non-Functional Requirements

REQ-NF-01 — The acquisition path shall be interrupt-assisted through ADC EOC handling; the SRS shall not claim a fixed ADC sampling frequency not established by the implementation.

REQ-NF-02 — The supplied implementation shall use statically allocated buffers and driver storage; no dynamic memory allocation is required by the logger design.

REQ-NF-03 — The implementation shall use the measured bus configurations of approximately 1 MHz SPI SCK and 375 kHz I2C SCL for the recorded setup.

REQ-NF-04 — New metadata records shall be validated using magic, checksum, sequence, and stored Flash-pointer information.

REQ-NF-05 — The startup path shall recover the latest valid metadata state after reset or power interruption by scanning both metadata sectors.

9. Verification and Test Plan

Requirement area	Test / Verification Method	Expected Evidence
ADC / EMA / NTC	SWD debug, variable inspection, temperature comparison	EOC updates raw/filtered values and a valid temperature is produced.
TIM3 timestamp	Observe timer ISR/system-seconds variable	System seconds increment at the configured 1 Hz interrupt rate under the stated clock assumption.
Ping-pong logging	RAM inspection + Flash trace	Buffer flag occurs at 64 samples; two 256-byte page writes are issued.
SPI Flash	Logic Analyzer + command/readback inspection	SCK ≈1 MHz; valid W25Q commands, addresses, data and chip-select timing.
I2C / OLED	Logic Analyzer + display observation	SCL ≈375 kHz; SSD1306 address 0x3C responds and framebuffer data is transferred.
Metadata recovery	Reset / power-cycle test	Newest valid sequence restores Flash pointers and system-time state.
Historical UI	Keypad operation + OLED observation	Recent/previous Flash windows, timestamp range and average temperature are displayed.
Alert system	Apply temperatures below 20°C, within 20–40°C, and above 40°C	Blue, Green, and Red LED states respectively; non-selected alert LEDs remain OFF.

10. Traceability Summary

System requirement	Primary implementation element	Verification evidence
Temperature acquisition	ADC1 + ADC EOC ISR + EMA + NTC calculation	SWD variables / temperature comparison
Timestamped logging	TIM3 + ping-pong buffers + logger	Timer state + RAM/Flash trace
Persistent storage	W25Q128 SPI driver + page program	Logic Analyzer + Flash inspection
Power-loss recovery	Dual-sector metadata journal	Reset/power-cycle recovery test
Historical viewing	Flash read + keypad + SSD1306	UI observation
Temperature alerts	PB12/PB13/PB14 state logic	Three threshold test cases

11. Implementation Constraints

- The project is a standalone bare-metal C implementation with direct peripheral-register access.
- SPI and I2C operations contain polling loops, so the complete application is not described as fully non-blocking.
- Measured bus frequencies recorded for this revision are SPI SCK ≈1 MHz and I2C SCL ≈375 kHz.
- The W25Q128 interface uses conventional SPI commands.

End of Software Requirements Specification — This document describes the implemented temperature logger architecture and its supported functional, interface, storage, alert, and verification requirements.